DESIGN, SIMULATION AND TECHNO-ECONOMIC FEASIBILITY ANALYSIS OF A GRID-CONNECTED INDUSTRIAL MICROGRID

Integrating Solar PV, Battery Energy Storage and Diesel Generation

UEE406 — Power System Practices
Engineering Report

Submitted By:

Rachit Saini

102304007 - 3D11

B.E. Electrical Engineering (IIIrd Year)



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Electrical & Instrumentation Engineering Department (EIED)

Index / Table of Contents

Index / Table of Contents 2

Abstract 4

Nomenclature 4

1. Introduction 5

    1.1 Objectives 5

    1.2 Scope 5

2. Design Basis and Assumptions 6

    2.1 Load-profile consistency 6

    2.2 Annual industrial energy 6

3. Industrial Load Assessment 7

4. Solar PV Sizing and Generation 8

    4.1 Daily and annual generation 8

    4.2 Direct PV utilization 9

5. Battery Energy Storage Sizing 10

    5.1 Energy capacity 10

    5.2 Battery state of charge 11

6. Diesel Generator Sizing 12

    6.1 Apparent power requirement 12

    6.2 Fuel estimate caveat 12

7. Microgrid Power Balance, Control and Outage Operation 13

    7.1 Simplified power balance 13

    7.2 Integrated normal operation 13

    7.3 Grid outage 15

    7.4 Required protection provisions 16

8. Annual Energy and Cost Comparison 17

9. Capital Cost and 20-Year Financial Model 18

    9.1 CAPEX assumptions 18

    9.2 Annual net cash flow 18

    9.3 NPV 18

9.4 IRR 18
9.5 Payback 19
10. Sensitivity Analysis 23
11. Environmental Assessment 28
12. Limitations and Risk Assessment 28
13. Recommendations 29
14. Conclusion 29
References and Data Sources 30
Appendix A — Industrial Load MATLAB Code 31
Appendix B — Solar PV MATLAB Code 32
Appendix C — Battery Sizing and SOC MATLAB Code 33
Appendix D — Diesel Generator Sizing MATLAB Code 34
Appendix E — Preliminary Microgrid Power-Balance MATLAB Code 35
Appendix F — Integrated Microgrid Normal-Operation MATLAB Code 36
Appendix G — Grid-Outage Simulation MATLAB Code 37
Appendix H — Annual Energy and Cost Comparison MATLAB Code 38
Appendix I — 20-Year Financial Analysis MATLAB Code 39
Appendix J — Sensitivity Analysis MATLAB Code 41

Abstract

This report presents a preliminary engineering and techno-economic assessment of a grid-connected industrial microgrid integrating solar photovoltaic (PV) generation, battery energy storage (BESS), and diesel backup generation. The representative facility is assumed to operate 16 hours/day for 300 days/year in Punjab, India. Candidate equipment comprises 350 kWp PV, 250 kWh BESS with a candidate 125 kW converter, and a 150 kVA diesel generator. MATLAB scripts model the industrial load, normalized solar production, battery state of charge, generator sizing, energy balance, outage scenarios, annual energy savings, 20-year cash flows, and sensitivity to financial assumptions. The hourly load series sums to 2,500 kWh/day and 750 MWh/year. The synthetic solar model estimates 511 MWh/year of generation, while the direct PV-load overlap model estimates approximately 375.1 MWh/year direct utilization. At an illustrative effective energy price of ₹8/kWh, gross direct-PV energy savings are approximately ₹30.01 lakh/year. An illustrative financial case gives initial CAPEX of ₹233.50 lakh, IRR of 7.51%, simple payback of 11.09 years, and NPV of approximately −₹37.44 lakh at a 10% discount rate. These outputs are not investment-grade estimates because the load, solar profile, tariffs, costs, degradation, and replacement assumptions require validation.

Keywords: industrial microgrid; solar PV; battery energy storage; diesel generator; load profile; energy management; NPV; IRR; payback.

Nomenclature

Symbol	Meaning	Unit
P_load	Industrial load power	kW
P_PV	PV output power	kW
E	Energy	kWh or MWh
H_PSH	Peak-sun-hours	h/day
PR	PV performance ratio	dimensionless
DoD	Battery depth of discharge	fraction or %
η	Efficiency	fraction or %
CAPEX	Capital expenditure	INR
OPEX	Operating expenditure	INR/year
CF_t	Incremental cash flow in year t	INR
r	Discount rate	fraction or %
NPV	Net present value	INR
IRR	Internal rate of return	%
EF_grid	Grid emissions factor	tCO_2/MWh

1. Introduction

Industrial electricity supply must support production loads while maintaining safety and continuity. A grid-connected microgrid combines the utility supply with local generation, storage, control, and backup resources. Solar PV can reduce grid-energy purchases, battery storage can provide short-duration support and shift some energy across time, and a diesel generator can extend backup duration when utility supply is unavailable.

1.1 Objectives

- Establish a representative industrial load profile and annual energy demand.
- Estimate candidate PV generation and direct PV utilization.
- Size battery energy for a 100 kW critical load and two-hour backup target.
- Estimate a candidate diesel-generator rating.
- Illustrate normal operation and grid-outage operation in MATLAB.
- Develop a preliminary cost and 20-year cash-flow model using NPV, IRR, and payback.
- Assess sensitivity and potential emissions reduction, and state limitations.

1.2 Scope

This work is a preliminary simulation and economic study, not a detailed electrical design. There are no measured factory interval data, site-specific irradiance series, final equipment quotations, verified generator fuel curves, or outage records in the current design basis. The calculations below distinguish assumptions from results.

2. Design Basis and Assumptions

Parameter	Value	Status
Location	Punjab, India	Representative location
Distribution basis	3-phase, 415 V, 50 Hz	LV basis only; actual site may be HT supplied
Peak hourly-average load	250 kW	Synthetic profile
Production schedule	16 h/day, 300 days/year	Assumption
Critical load	100 kW	Assumption
Backup target	2 h	Design target
PV candidate	350 kWp	Preliminary
BESS candidate	250 kWh	Preliminary
Battery converter	125 kW	Candidate; verify transients/reactive power
Diesel generator	150 kVA	Candidate; verify starting/derating
Project life	20 years	Financial assumption
Effective electricity price	₹8/kWh	Illustrative, not actual tariff
Solar assumption	5 PSH/day; PR = 0.80	Illustrative, not site-validated
Discount rate	10%	Illustrative financial assumption

2.1 Load-profile consistency

The 16 hourly values used in the model are [100, 120, 140, 160, 180, 200, 160, 120, 130, 150, 170, 190, 250, 200, 120, 110] kW. Their sum is 2,500, not 2,400. Therefore the calculations in this report use 2,500 kWh/day and 750 MWh/year. This correction is important because annual energy directly affects estimated PV savings and the financial model.

2.2 Annual industrial energy

For hourly intervals, energy is the sum of power multiplied by the interval duration:

$E_{daily} = \Sigma(P_{load,t} \times \Delta t) = 2{,}500$ kWh/day, where $\Delta t = 1$ h.

$E_{annual} = E_{daily} \times N_{operating_days} = 2{,}500 \times 300 = 750{,}000$ kWh/year = 750 MWh/year.

Average load over the 16-hour operating period = 2,500/16 = 156.25 kW. The peak hourly-average value is 250 kW. Standby consumption outside the production period is excluded.

3. Industrial Load Assessment

The synthetic profile represents a varying manufacturing load rather than a constant 150 kW demand. It is used for hourly overlap with PV generation and for illustrating grid import. Actual design should use measured 15-minute or 30-minute interval data, maximum demand, power factor, motor starting, shift changes, and weekend/holiday schedules.

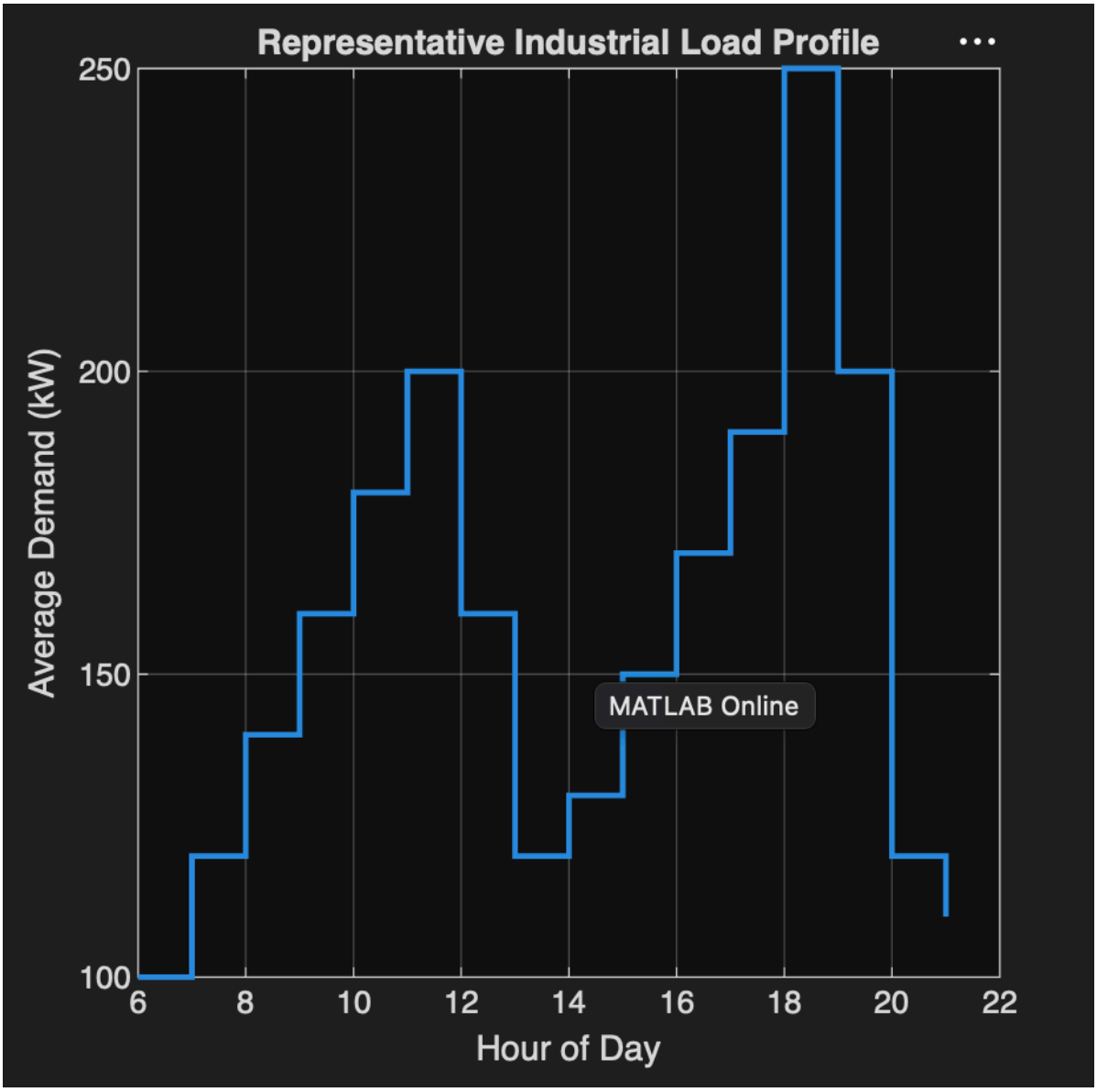


Figure 1.1. Representative industrial load profile.

Metric	Calculation	Result
Daily energy	Σ P_t × 1 h	2,500 kWh/day
Annual energy	2,500 × 300	750 MWh/year
Average operating load	2,500/16	156.25 kW
Peak hourly-average load	max(P_t)	250 kW
Critical load target	Assumption	100 kW

4. Solar PV Sizing and Generation

4.1 Daily and annual generation

A preliminary 350 kWp array is estimated using five peak-sun-hours per day and a performance ratio of 0.80:

$E_{PV,daily} = P_{PV} \times H_{PSH} \times PR = 350\ kWp \times 5\ h/day \times 0.80 = 1{,}400\ kWh/day.$

$E_{PV,annual} = 1{,}400 \times 365 = 511{,}000\ kWh/year = 511\ MWh/year.$

This is an energy-yield estimate using an assumed constant daily average, not a validated site-specific yield model. Seasonal variation, shading, temperature, orientation, soiling, inverter clipping, and downtime are not resolved.

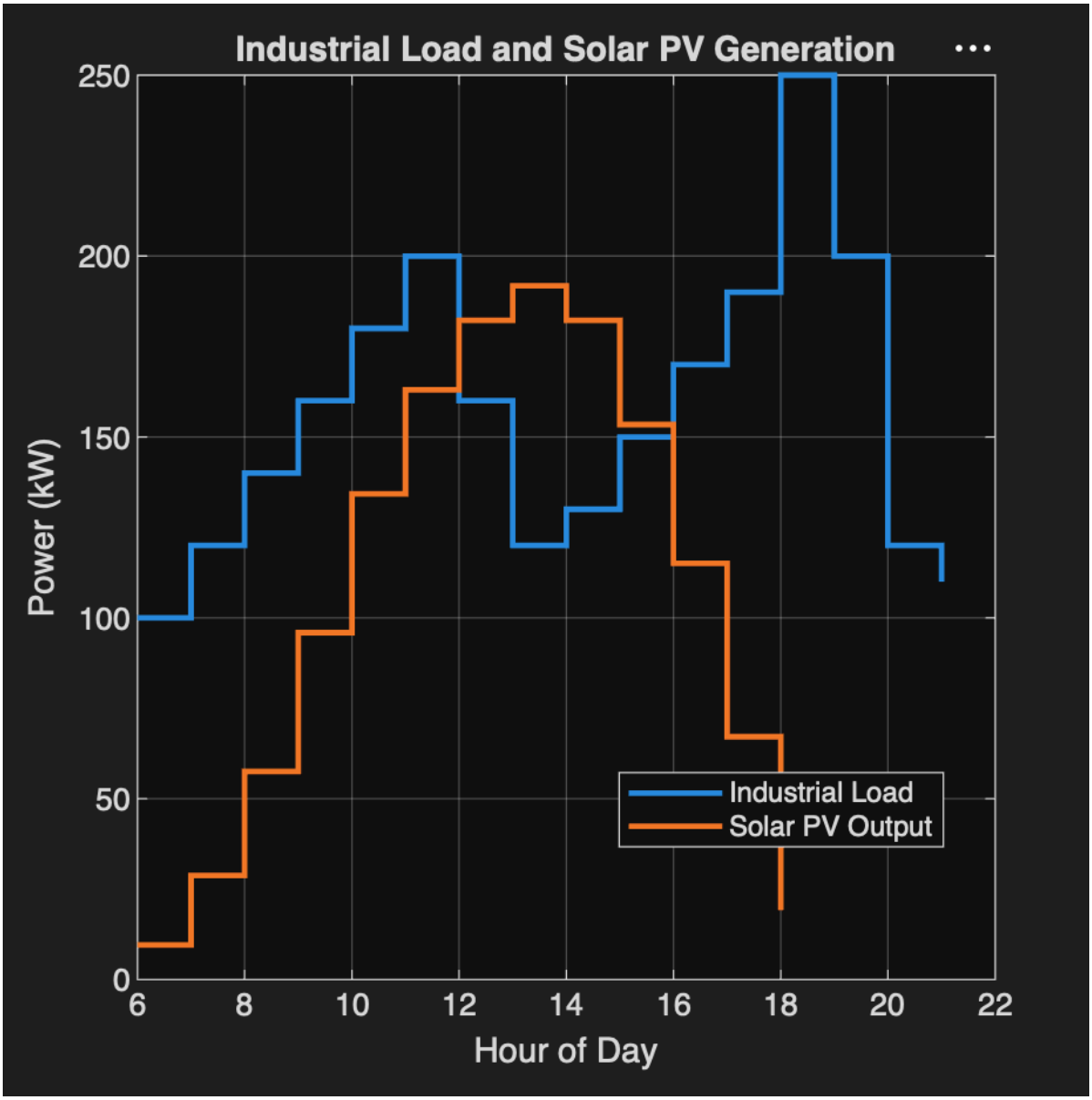


Figure 2.1. Industrial load and illustrative solar PV generation.

4.2 Direct PV utilization

For each modelled hour, direct PV use is the lower of PV power and load power:

$E_{PV,direct,day} = \Sigma \min(P_{PV,t}, P_{load,t}) \times \Delta t.$

Using the normalized synthetic curve, the calculated direct overlap is approximately 1,250.4 kWh per operating day. Multiplying by 300 operating days gives approximately 375.1 MWh/year. This is not the same as total PV generation; potential surplus, off-shift production, battery losses, and exports require separate accounting.

The generation-to-load ratio 511/750 ≈ 68.1% is only a ratio of annual energy estimates. It is not the actual fraction of factory consumption supplied by PV. Direct utilization in this illustrative model is approximately 375.1/750 ≈ 50.0% of annual load.

5. Battery Energy Storage Sizing

5.1 Energy capacity

The target is to supply 100 kW of critical load for two hours. The useful AC energy requirement is:

$E_{required} = P_{critical} \times t_{backup} = 100\ kW \times 2\ h = 200\ kWh.$

Accounting for depth of discharge and discharge efficiency:

$E_{nominal} \geq E_{required}/(DoD \times \eta_{discharge}) = 200/(0.90 \times 0.95) = 233.92\ kWh.$

The candidate 250 kWh battery gives an idealized AC-equivalent energy of $250 \times 0.90 \times 0.95 = 213.75$ kWh, corresponding to about 2.14 h at 100 kW when new and fully charged. This small margin does not establish two-hour autonomy at end of life.

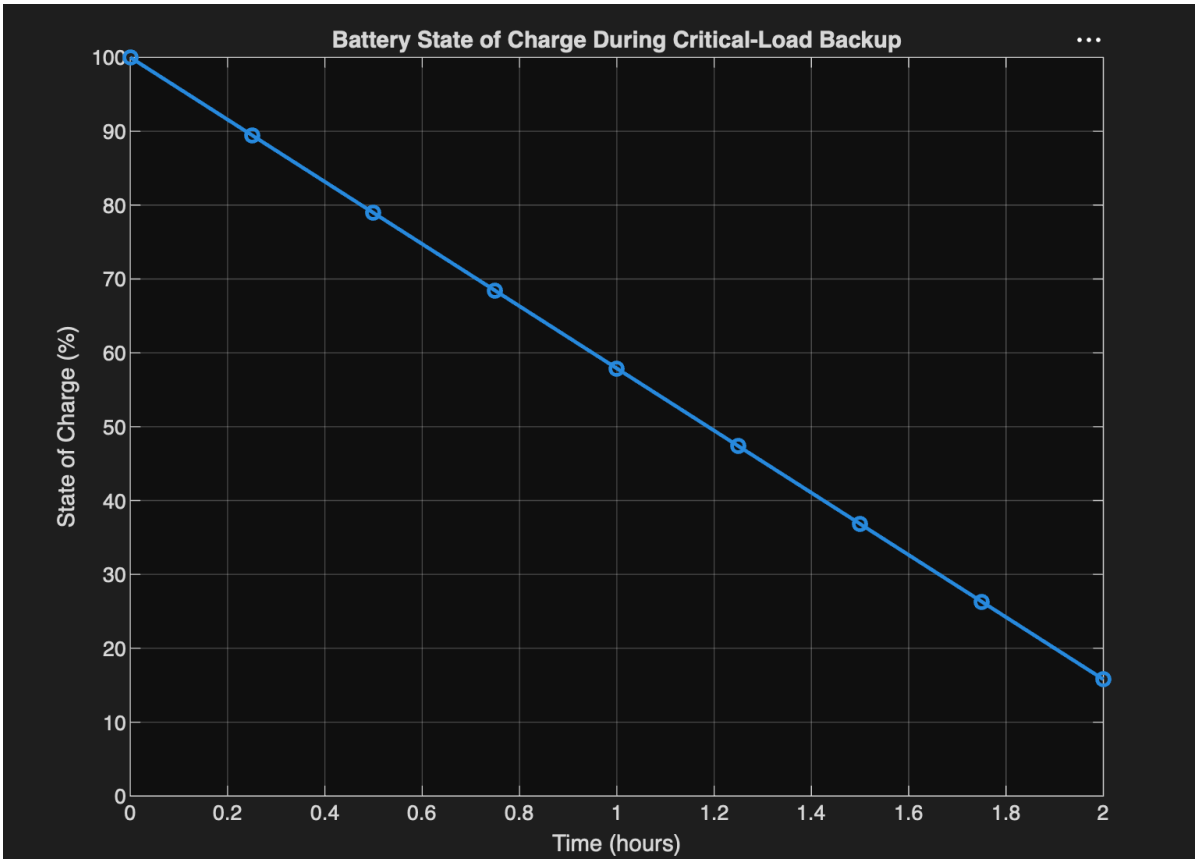


Figure 3.1. Battery state of charge during critical-load backup.

5.2 Battery state of charge

A simplified discharge model can be written as $SOC(t+\Delta t) = SOC(t) - [P_{batt} \times \Delta t/(E_{nominal} \times \eta_{discharge})] \times 100\%$, subject to minimum SOC and power constraints. The supplied MATLAB plot assumes a full initial SOC and discharges toward a minimum reserve. It is an energy-only model and does not represent electrochemical dynamics, temperature, voltage limits, or battery aging.

BESS parameter	Assumption / result
Nominal capacity	250 kWh
Critical load	100 kW
Backup duration target	2 h
Depth of discharge	90%
Discharge efficiency	95%
Idealized AC-equivalent energy	213.75 kWh
Idealized duration at 100 kW	2.14 h
Converter candidate	125 kW

6. Diesel Generator Sizing

6.1 Apparent power requirement

For a critical real-power load P = 100 kW and assumed power factor PF = 0.90:

S_load = P/PF = 100/0.90 = 111.11 kVA.

If a preliminary design target limits normal generator loading to 80%, then:

S_generator,min = S_load/0.80 = 111.11/0.80 = 138.89 kVA.

A 150 kVA set is therefore a candidate. At PF 0.90 its nominal real-power capability is 150 × 0.90 = 135 kW. The 100 kW steady load corresponds to 100/135 × 100 ≈ 74.1% of that nominal real-power capability.

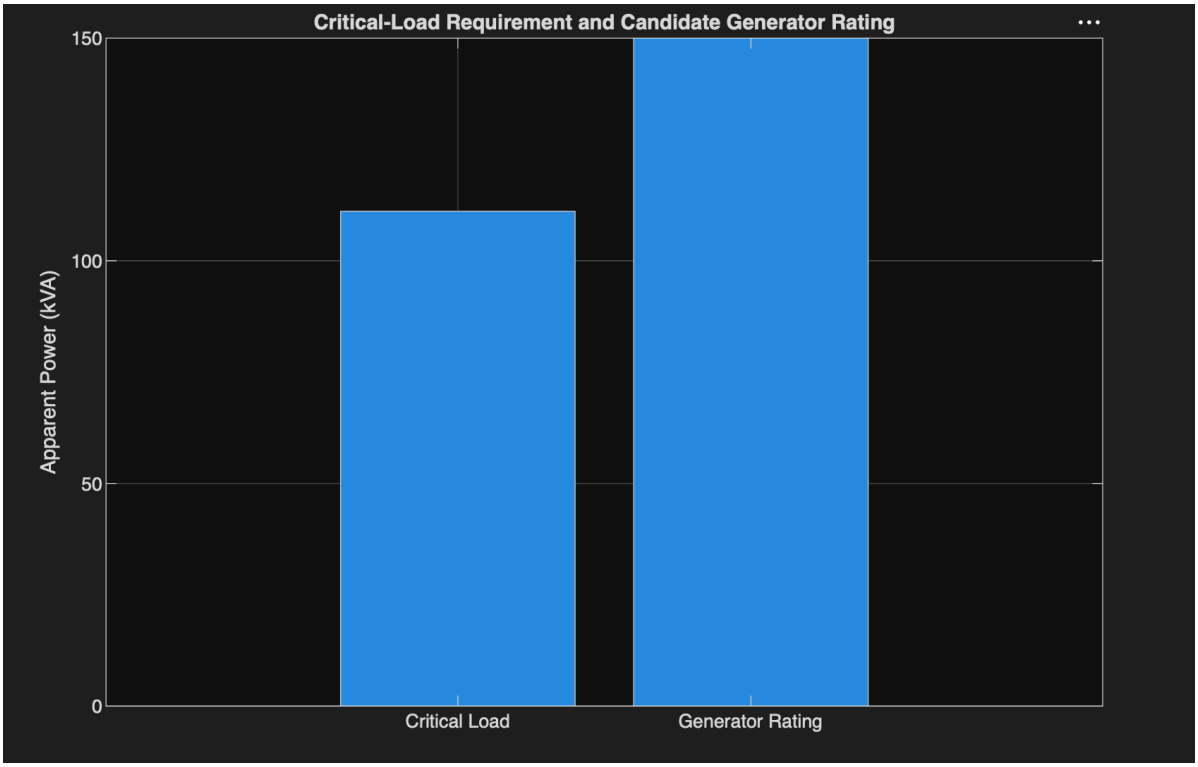


Figure 4.1. Critical-load requirement and candidate generator rating.

6.2 Fuel estimate caveat

An illustrative fuel factor of 0.27 L/kWh would imply 100 kW × 2 h × 0.27 = 54 L for a two-hour 100 kW generator run. This factor is not verified manufacturer data and must not be treated as a procurement or operating estimate. Actual fuel use depends on generator model and loading.

7. Microgrid Power Balance, Control and Outage Operation

7.1 Simplified power balance

For each time step, a simplified grid-connected energy balance is P_load = P_PV,direct + P_BESS,discharge + P_grid,import − P_BESS,charge − P_export, subject to sign convention and conversion losses. In the preliminary normal-operation script, PV serves load first, surplus PV charges the battery where available, and the grid supplies remaining deficit. The current simplified case does not claim detailed export or tariff optimization.

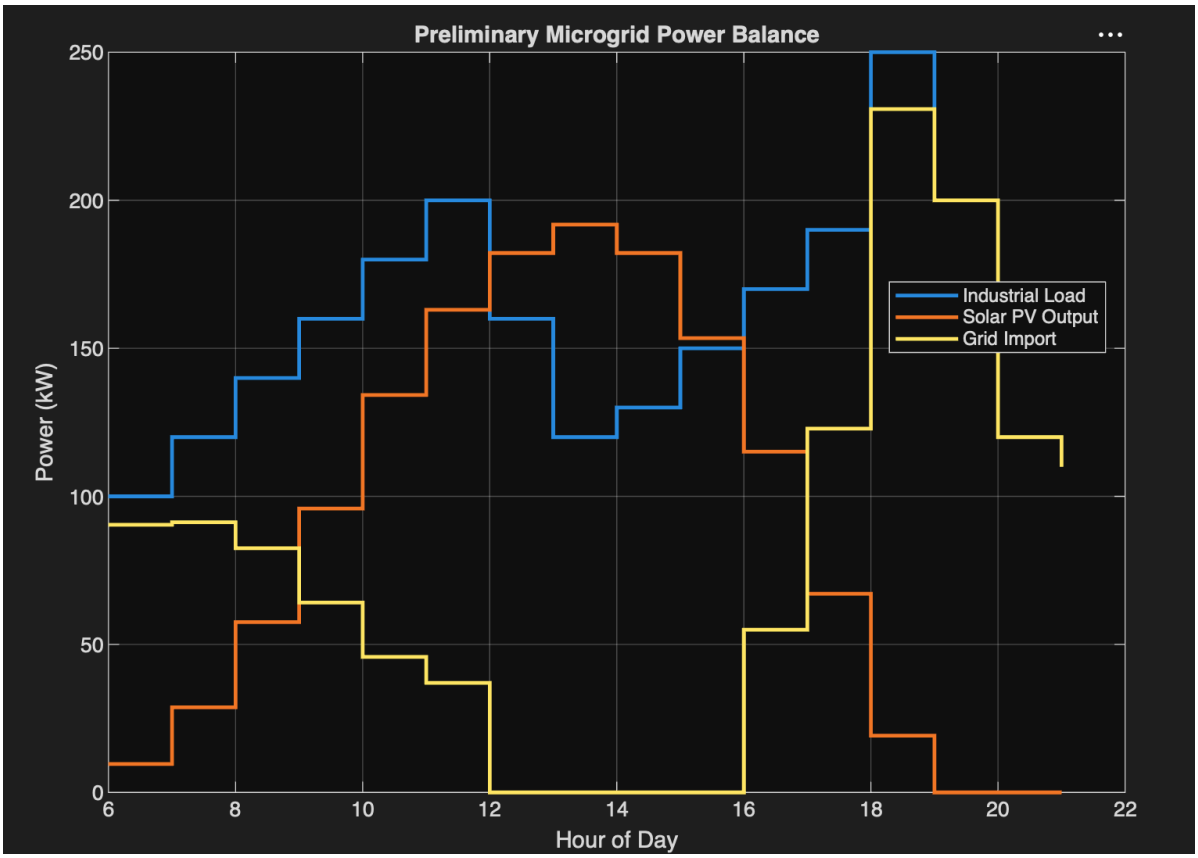


Figure 5.1. Preliminary microgrid power balance.

7.2 Integrated normal operation

The integrated plots compare load and PV generation, grid import, and battery SOC. They illustrate an energy-management concept rather than a validated controller. Battery discharge is restricted in the normal-operation example to preserve backup reserve.

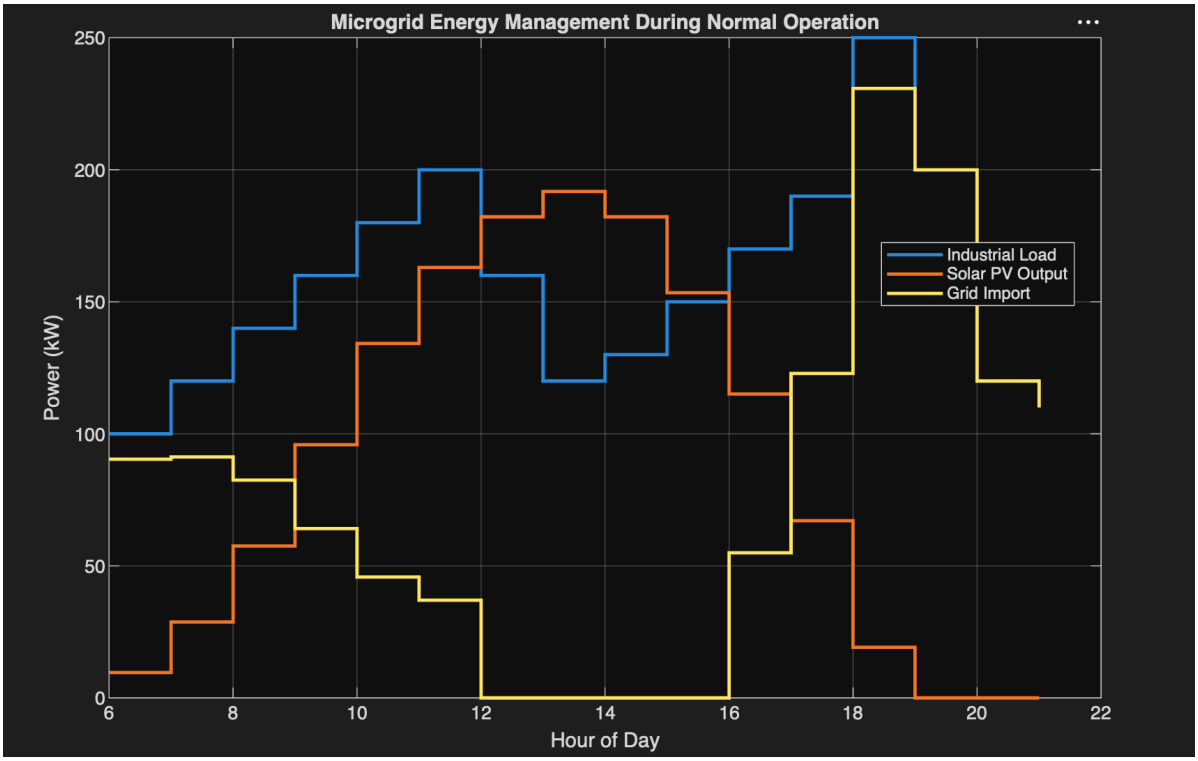


Figure 6.1. Microgrid energy management during normal operation.

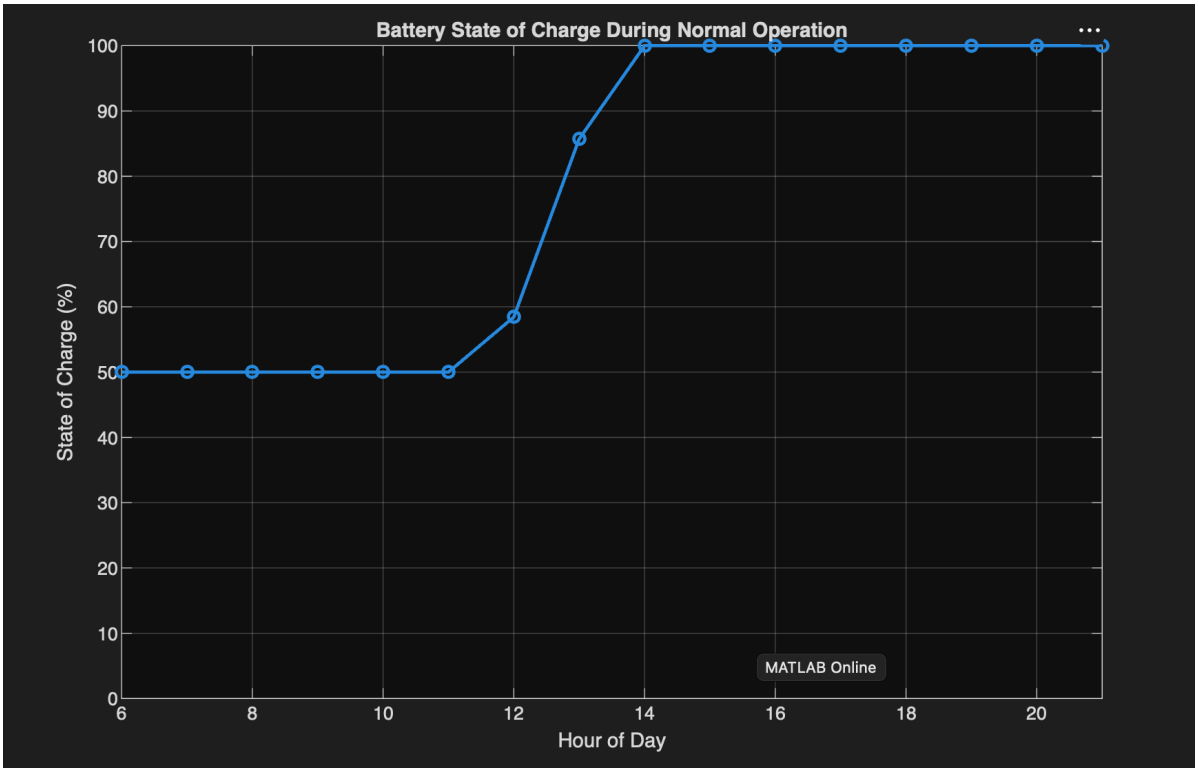


Figure 6.2. Battery state of charge during normal operation.

7.3 Grid outage

During an outage, the utility point of connection must be isolated before an islanded bus is energized. The battery-only scenario estimates critical-load SOC depletion over the backup interval. A second illustrative scenario includes a 0.25-hour generator-start delay and a subsequent diesel supply interval. The plots do not model switching transients, ramp rates, voltage/frequency stability, synchronization, or detailed protection.

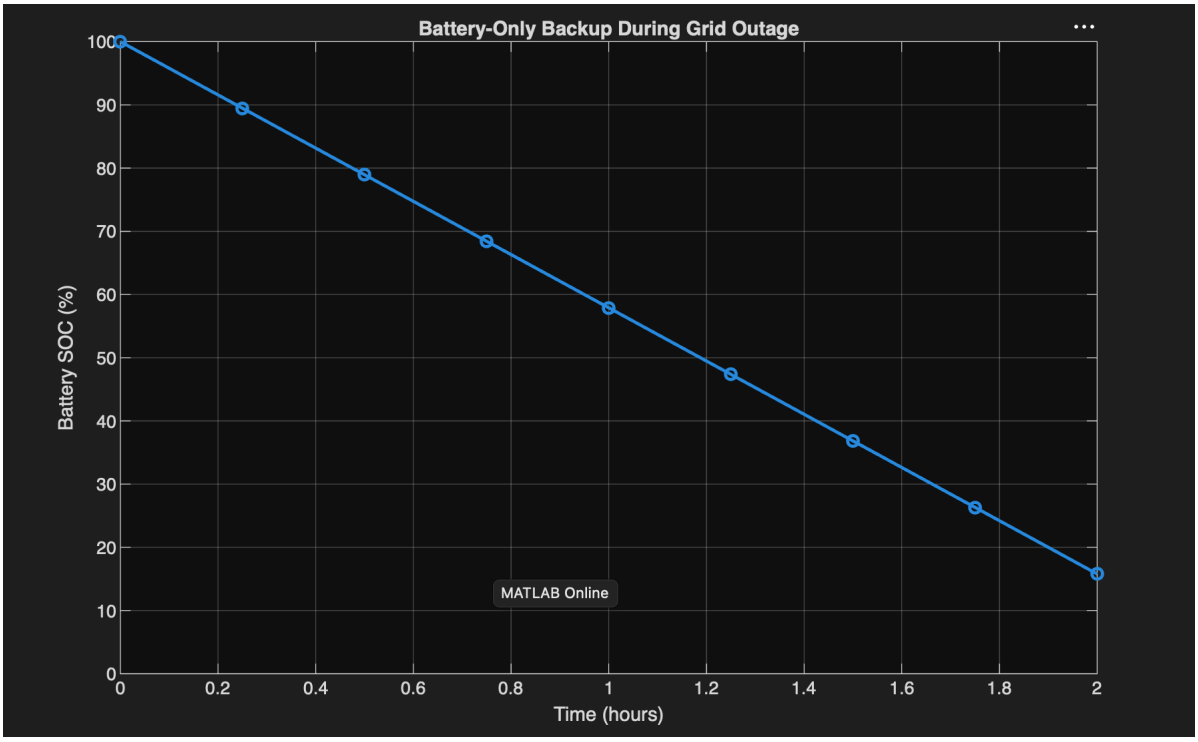


Figure 7.1. Battery-only backup during grid outage.

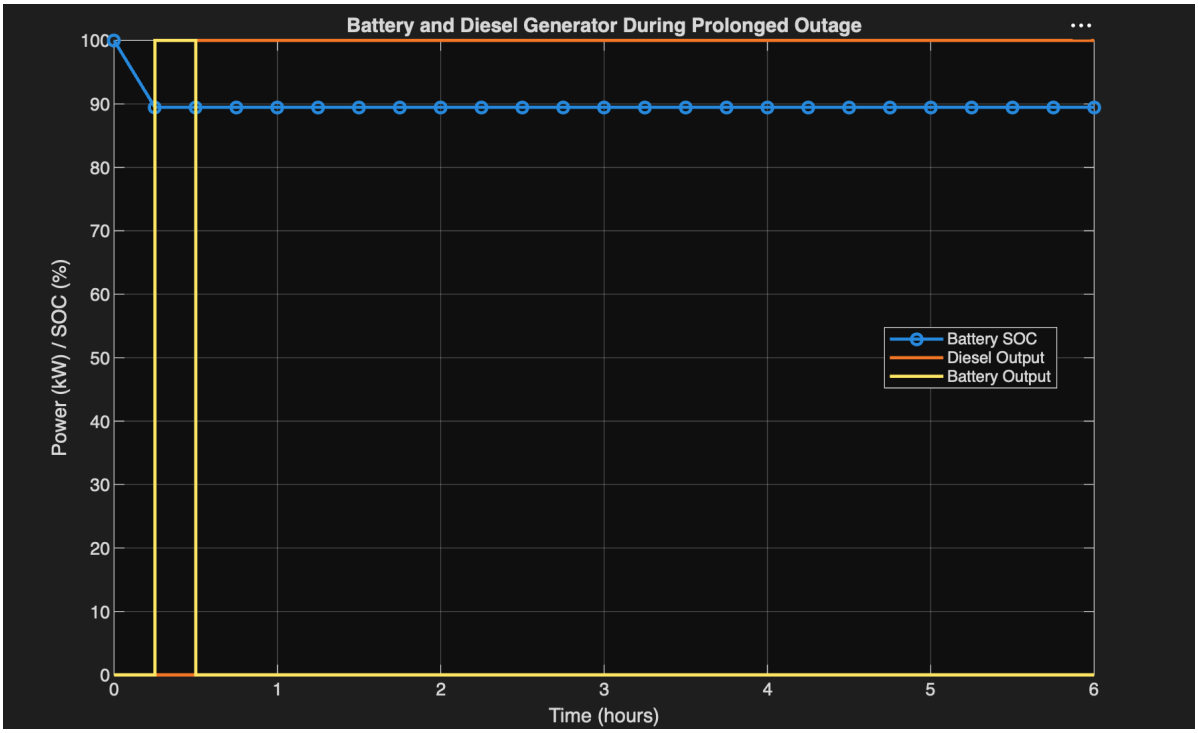


Figure 7.2. Battery and diesel generator during prolonged outage.

7.4 Required protection provisions

- Utility isolation and anti-islanding protection.
- Grid-forming capability and black-start strategy for islanded operation.
- Overcurrent, earth-fault, under/overvoltage, under/overfrequency, and synchronization protection as applicable.
- Interlocking to prevent unintended parallel operation with the utility.
- Critical-load prioritization and load shedding.
- Coordination among PV inverter, BESS converter, diesel controller, and site protection.

8. Annual Energy and Cost Comparison

The baseline grid-energy cost is estimated using an illustrative effective price T = ₹8/kWh. This is not the verified tariff for the site.

C_baseline = E_annual × T = 750,000 kWh/year × ₹8/kWh = ₹60,00,000/year = ₹60.00 lakh/year.

Direct PV saving = E_PV,direct × T = 375,123 kWh/year × ₹8/kWh ≈ ₹30.01 lakh/year.

Grid imports after direct PV use = 750,000 − 375,123 = 374,877 kWh/year ≈ 374.9 MWh/year. Illustrative post-PV energy cost = 374,877 × ₹8 ≈ ₹29.99 lakh/year.

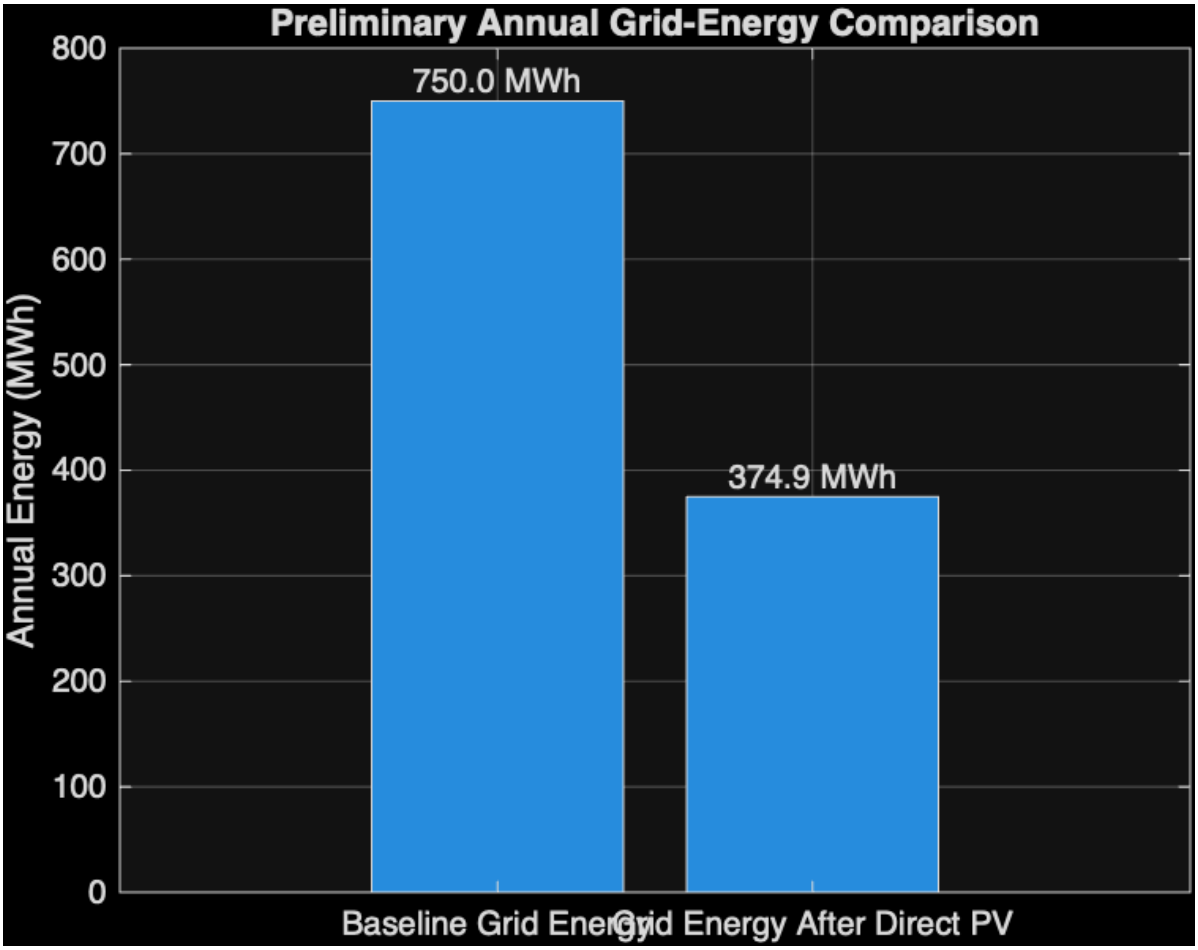


Figure 8.1. Preliminary annual grid-energy comparison.

Metric	Baseline	After direct PV use
Annual load	750.0 MWh	750.0 MWh
Direct PV use	0 MWh	375.1 MWh
Grid imports	750.0 MWh	374.9 MWh
Illustrative energy cost	₹60.00 lakh/year	₹29.99 lakh/year
Gross direct-PV saving	—	₹30.01 lakh/year

This comparison excludes fixed charges, demand charges, taxes, O&M, battery-mediated savings, export compensation, and any change in diesel operation. In addition, the exact industrial tariff category and billing units must be checked. A charge stated in ₹/kVAh should not automatically be treated as ₹/kWh.

9. Capital Cost and 20-Year Financial Model

9.1 CAPEX assumptions

Component	Illustrative amount	Basis
PV 350 kWp	₹126.00 lakh	Lower-end planning benchmark used in the model
BESS 250 kWh	₹62.50 lakh	Lower-end planning benchmark used in the model
150 kVA generator	₹15.00 lakh	Assumption; quotation required
Integration	₹20.00 lakh	Assumption; scope to verify
Other costs	₹10.00 lakh	Assumption
Total CAPEX	₹233.50 lakh	Sum of above

C_0 = C_PV + C_BESS + C_DG + C_integration + C_other = 126 + 62.5 + 15 + 20 + 10 = ₹233.5 lakh. The PV and BESS costs are indicative only; scope must be checked for inclusions to prevent double-counting.

9.2 Annual net cash flow

Gross direct-PV savings are approximately ₹30.01 lakh/year. With illustrative OPEX of ₹5 lakh/year and no incremental fuel cost included, annual net cash flow before replacement is 30.01 − 5.00 = ₹25.01 lakh/year. This is a simplified incremental cash-flow proxy, not a complete operational cash flow.

9.3 NPV

NPV = $-C_0 + \Sigma[t=1 \text{ to } N]\ CF_t/(1+r)^t$, where C_0 is initial CAPEX, CF_t is the incremental cash flow in year t, r is the discount rate, and N is the project life. Positive NPV indicates value creation relative to the chosen discount rate, subject to the validity of inputs.

9.4 IRR

IRR is the rate i that satisfies $0 = -C_0 + \Sigma[t=1 \text{ to } N]\ CF_t/(1+i)^t$. It is interpreted alongside NPV, especially when replacements create additional negative cash flows.

9.5 Payback

Simple payback is the time until cumulative undiscounted cash flow recovers C_0. Discounted payback uses $CF_t/(1+r)^t$ instead. Simple payback ignores time value of money; discounted payback may not occur within the project horizon even when simple payback does.

```
==========================================
 INDUSTRIAL MICROGRID FINANCIAL ANALYSIS
==========================================
Project life                 : 20 years
Discount rate                : 10.00%
PV capital cost              : Rs 126.00 lakh
Battery capital cost         : Rs 62.50 lakh
DG capital cost              : Rs 15.00 lakh
Integration capital cost     : Rs 20.00 lakh
Other capital cost           : Rs 10.00 lakh
------------------------------------------
Total initial CAPEX           : Rs 233.50 lakh
Annual gross PV saving        : Rs 30.01 lakh
Annual operating expenses     : Rs 5.00 lakh
Annual net cash flow before replacement: Rs 25.01 lakh
------------------------------------------
NPV                          : Rs -3
IRR                          : 7.51%
Simple payback               : 11.09 years
Discounted payback           : Not reached within project life
==========================================
```

MATLAB Online

Financial model console output showing assumptions and calculated indicators.

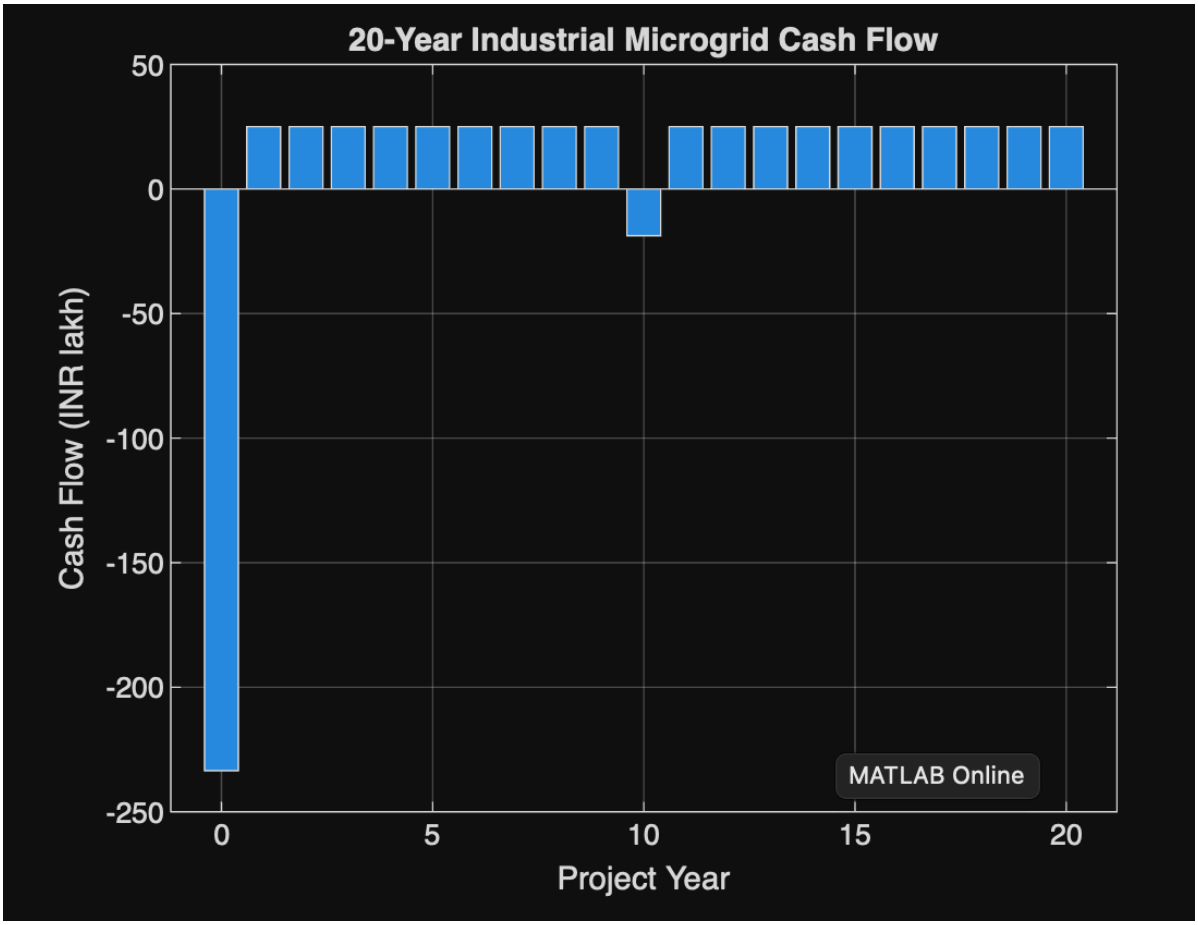


Figure 8.2. Annual cash flow over the 20-year project horizon.

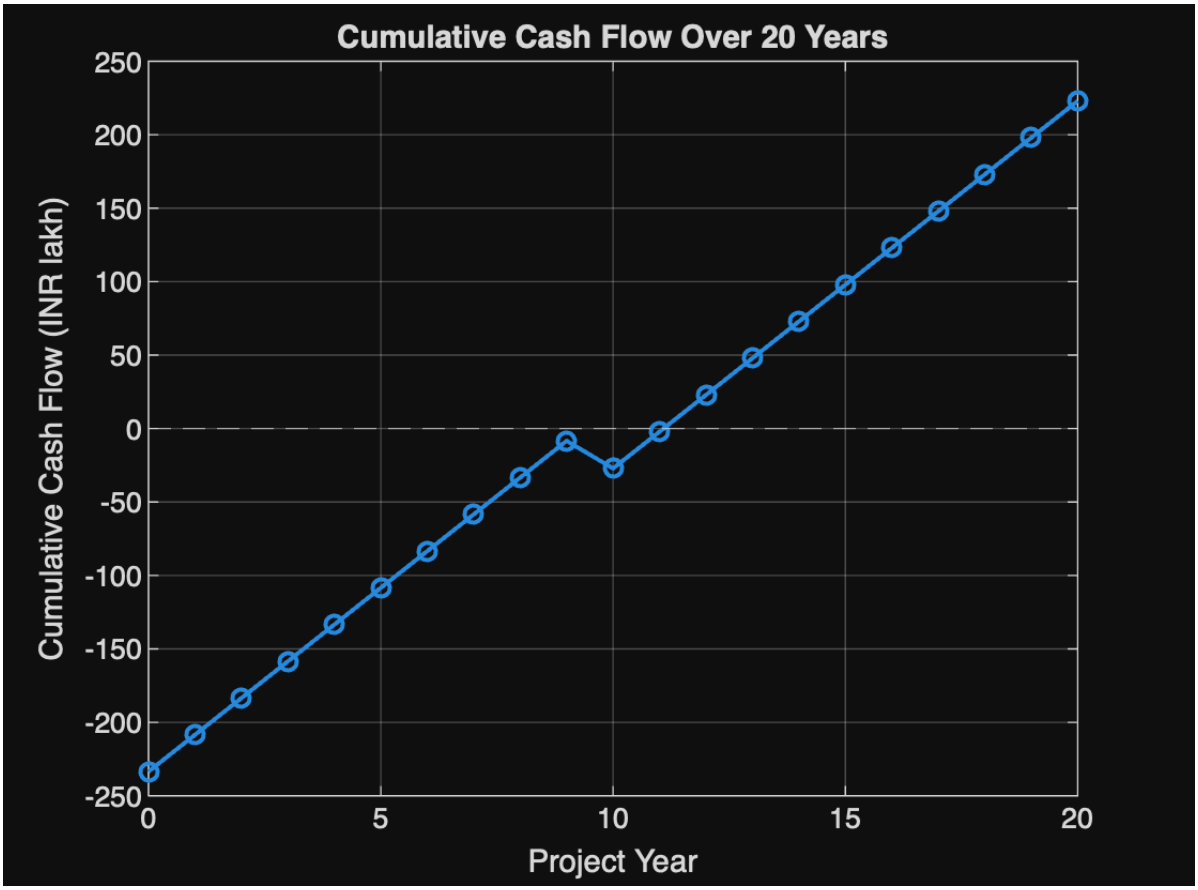


Figure 8.3. Cumulative cash flow over 20 years.

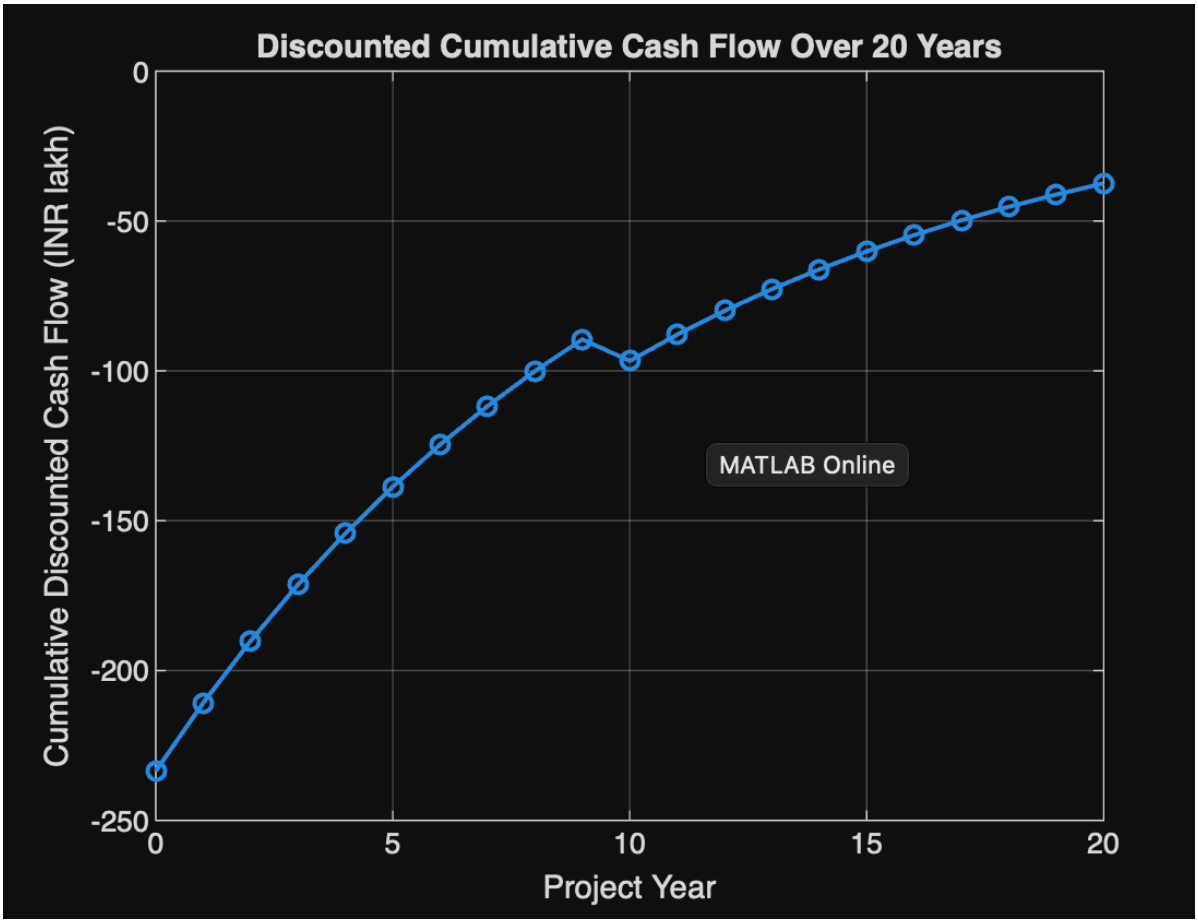


Figure 8.4. Discounted cumulative cash flow over 20 years.

Financial indicator	Model output	Interpretation
Initial CAPEX	₹233.50 lakh	Illustrative cost case
Gross PV saving	₹30.01 lakh/year	Before operating and replacement costs
OPEX	₹5.00 lakh/year	Illustrative assumption
Net annual cash flow before replacement	₹25.01 lakh/year	Simplified
IRR	7.51%	Below 10% discount rate
Simple payback	11.09 years	Undiscounted model output
Discounted payback	Not reached within 20 years	At 10% discount rate
NPV at 10%	Approx. –₹37.44 lakh	Negative base case

Interpretation: the illustrative base case does not meet a 10% return requirement because IRR is below 10% and NPV is negative. Actual results may change materially after verified tariffs, quotations, degradation, battery replacement, diesel operating costs, and bill components are incorporated.

10. Sensitivity Analysis

One-factor-at-a-time sensitivity tests change one assumption while holding others fixed. They show direction and approximate magnitude, not a probability distribution or a full combined scenario.

```
==========================================
 INDUSTRIAL MICROGRID SENSITIVITY ANALYSIS
==========================================
Base-case CAPEX: Rs 233.50 lakh
Annual PV saving: Rs 30.01 lakh
Annual OPEX: Rs 5.00 lakh
Base-case NPV: Rs -37.44 lakh
```

Sensitivity-analysis console output: base case and summary.

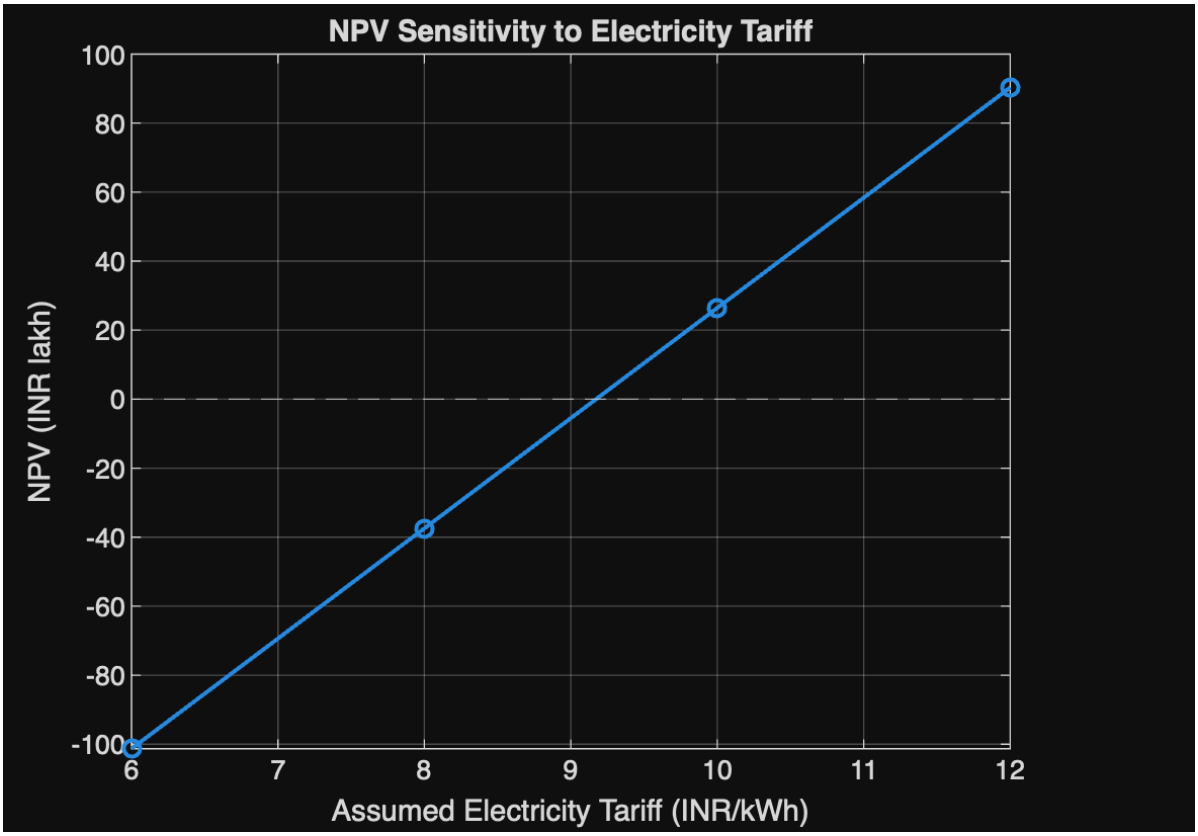


Figure 8.5. NPV sensitivity to assumed electricity tariff.

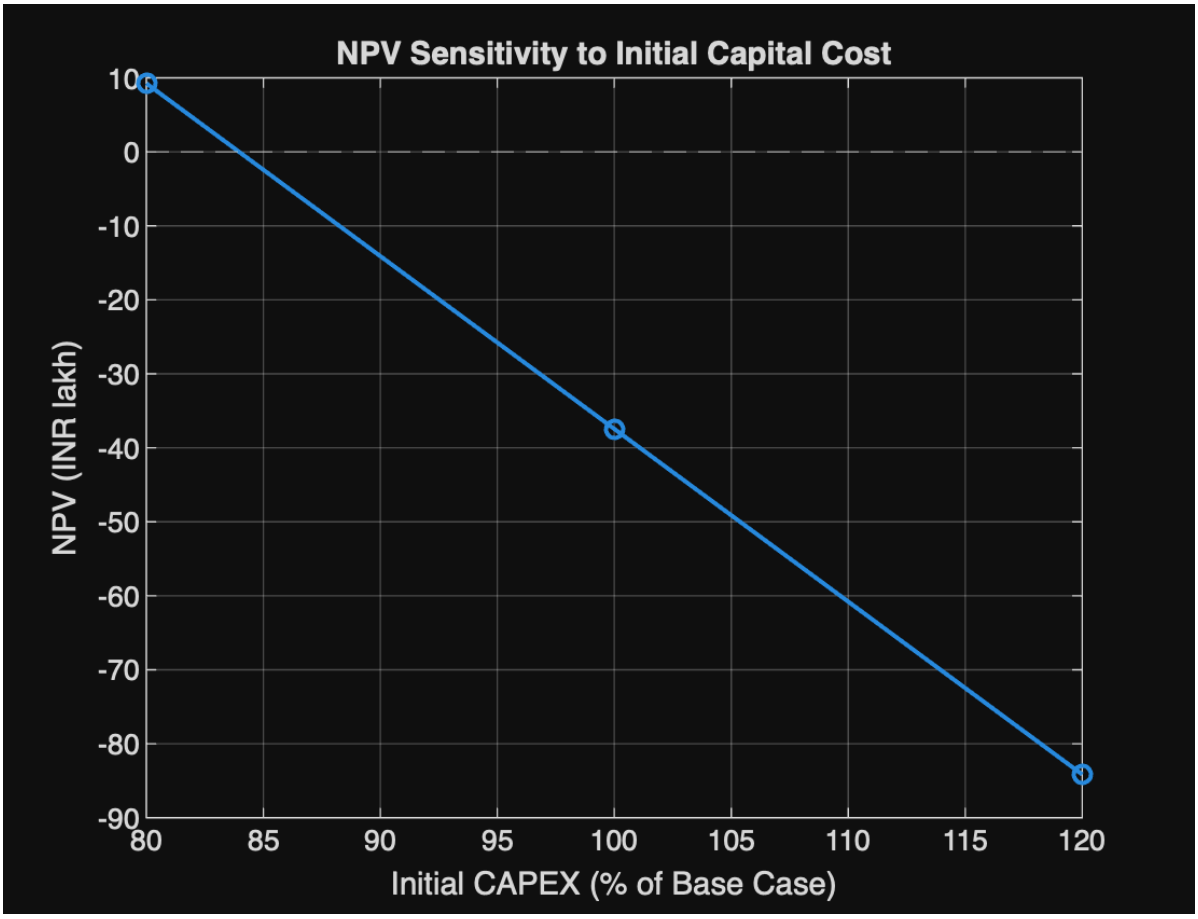


Figure 8.6. NPV sensitivity to initial capital cost.

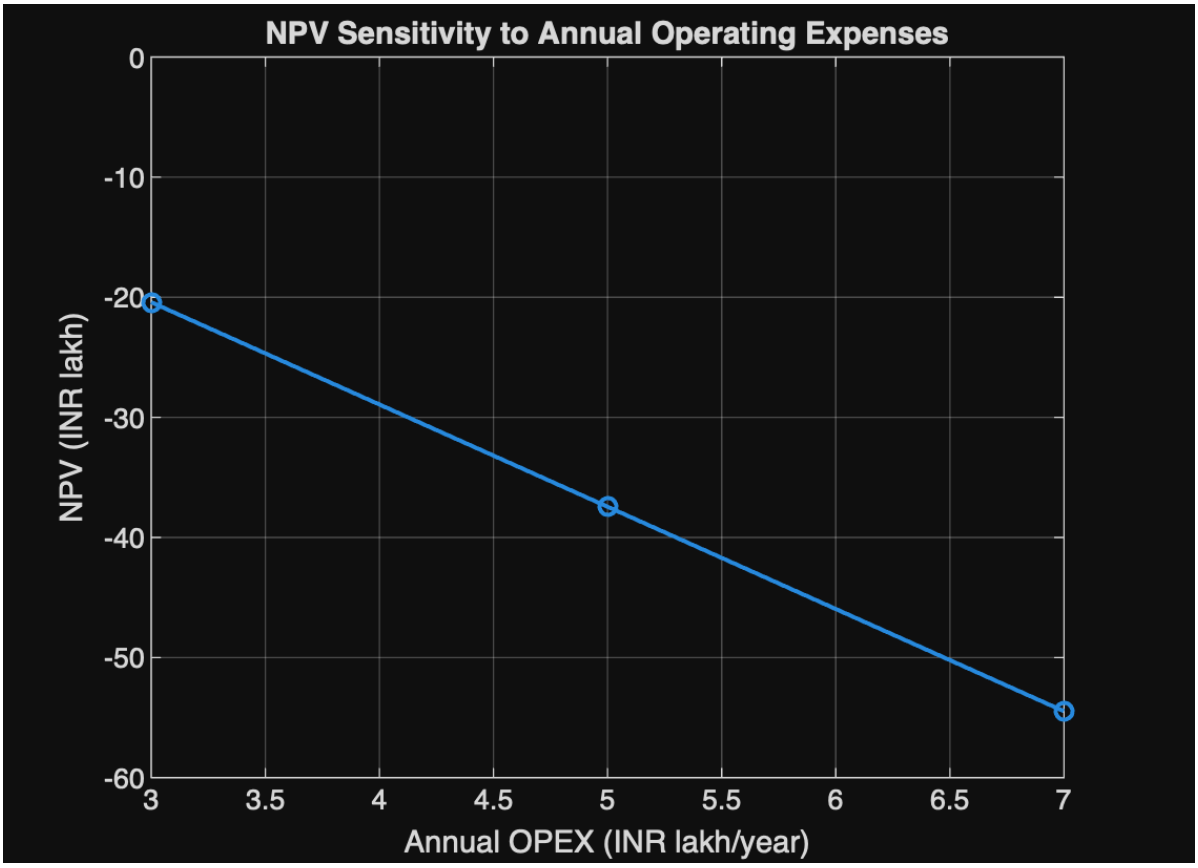


Figure 8.7. NPV sensitivity to annual operating expenses.

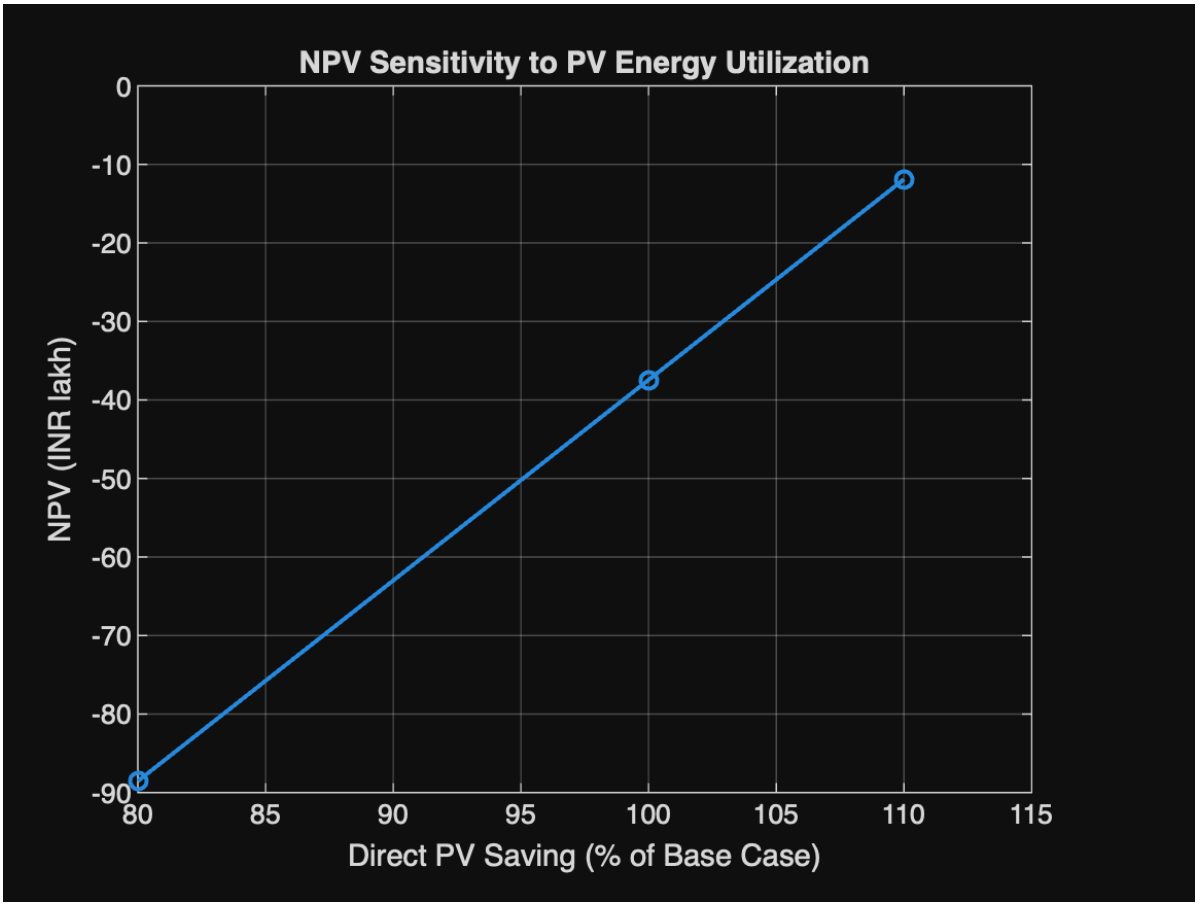


Figure 8.8. NPV sensitivity to direct PV savings.

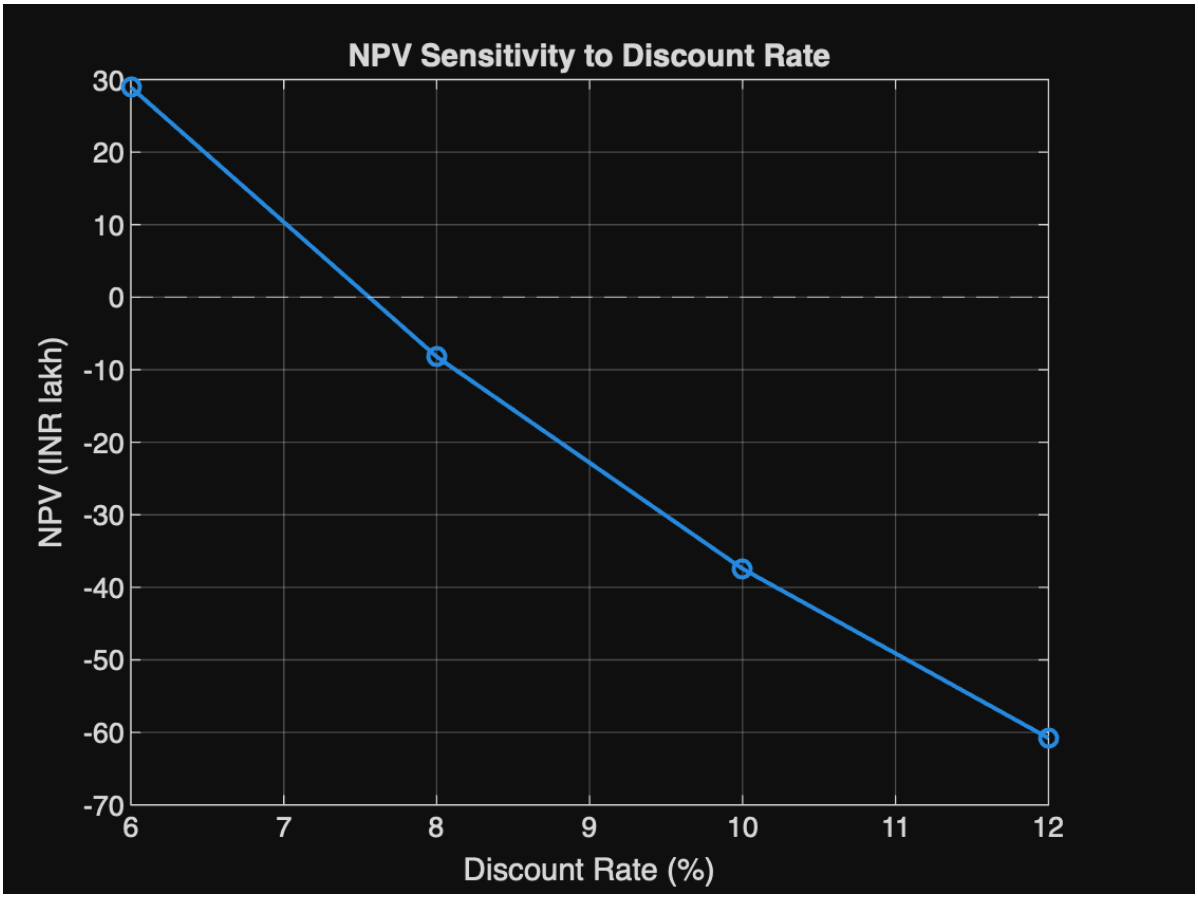


Figure 8.9. NPV sensitivity to discount rate.

```
--- TARIFF SENSITIVITY ---

Tariff Rs 6.00/kWh: NPV = Rs -101.32 lakh
Tariff Rs 8.00/kWh: NPV = Rs -37.44 lakh
Tariff Rs 10.00/kWh: NPV = Rs 26.43 lakh
Tariff Rs 12.00/kWh: NPV = Rs 90.30 lakh

--- CAPEX SENSITIVITY ---

CAPEX 80% of base: NPV = Rs 9.26 lakh
CAPEX 100% of base: NPV = Rs -37.44 lakh
CAPEX 120% of base: NPV = Rs -84.14 lakh

--- OPEX SENSITIVITY ---

OPEX Rs 3.00 lakh/year: NPV = Rs -20.42 lakh
OPEX Rs 5.00 lakh/year: NPV = Rs -37.44 lakh
OPEX Rs 7.00 lakh/year: NPV = Rs -54.47 lakh

--- PV GENERATION SENSITIVITY ---

PV saving 80% of base: NPV = Rs -88.54 lakh
PV saving 100% of base: NPV = Rs -37.44 lakh
PV saving 110% of base: NPV = Rs -11.89 lakh

--- DISCOUNT RATE SENSITIVITY ---

Discount rate 6%: NPV = Rs 28.93 lakh
Discount rate 8%: NPV = Rs -8.21 lakh
Discount rate 10%: NPV = Rs -37.44 lakh
Discount rate 12%: NPV = Rs -60.78 lakh

Sensitivity analysis completed.
```

Sensitivity-analysis console output: scenario NPV results.

Variable	Scenario	NPV (₹ lakh)
Tariff	₹6/kWh	−101.32
Tariff	₹8/kWh	−37.44
Tariff	₹10/kWh	+26.43
Tariff	₹12/kWh	+90.30
CAPEX	80% of base	+9.26
CAPEX	100% of base	−37.44
CAPEX	120% of base	−84.14
OPEX	₹3 lakh/year	−20.42
OPEX	₹5 lakh/year	−37.44

OPEX	₹7 lakh/year	−54.47
PV savings	80% of base	−88.54
PV savings	100% of base	−37.44
PV savings	110% of base	−11.89
Discount rate	6%	+28.93
Discount rate	8%	−8.21
Discount rate	10%	−37.44
Discount rate	12%	−60.78

The model indicates strong sensitivity to effective electricity price and capital cost. Even at 110% of base PV savings, NPV remains negative in the reported scenario. The results must be recalculated when the cash-flow model is updated with realistic annual PV degradation, tariff escalation, replacement, and diesel costs.

11. Environmental Assessment

A preliminary emissions estimate uses direct PV utilization of 375.1 MWh/year and an illustrative grid emission factor of 0.70 tCO_2/MWh:

CO_2 avoided = E_PV,direct × EF_grid = 375.1 MWh/year × 0.70 tCO_2/MWh ≈ 262.6 tCO_2/year.

A simple 20-year multiplication gives 262.6 × 20 ≈ 5,252 tCO_2. This assumes constant performance and a constant grid factor. It excludes PV degradation, changes in grid emissions, manufacturing and end-of-life emissions, battery replacement, and diesel-generator emissions. Use a documented grid factor in the final submitted version.

Net avoided emissions should be calculated as grid emissions displaced minus diesel emissions attributable to the proposed operating strategy, with the baseline backup arrangement also considered.

12. Limitations and Risk Assessment

- Load profile is synthetic; actual maximum demand and connected-load schedule need reconciliation.
- Solar curve is synthetic and normalized; no site-specific irradiance simulation is included.
- PV direct-use calculation covers the synthetic operating-hour overlap and does not fully resolve off-shift generation/export.
- Battery model is simplified and does not simulate aging, thermal behaviour, voltage, or power limits in detail.
- Outage plots are illustrative and do not establish dynamic stability or protection performance.
- Diesel fuel consumption, annual outage hours, and generator pricing are not verified.
- Tariff input is an assumed effective ₹/kWh and is not a reproduction of the facility's complete bill.

- CAPEX and OPEX require supplier quotations and a scope review to prevent omissions or double-counting.
- Financial outputs are conditional and should not be treated as bankable estimates.
- Environmental result uses an illustrative emission factor and is not a lifecycle assessment.

13. Recommendations

- Obtain interval load data and confirm the critical-load inventory.
- Validate the PV yield with site-specific weather/irradiance data and system design.
- Obtain quotations and datasheets for PV, BESS, generator, EMS, converters, protection, installation, and commissioning.
- Use actual utility bills to model energy charges, fixed charges, demand charges, power factor treatment, and export settlement.
- Add PV degradation, battery degradation, replacement timing, tariff escalation, and verified diesel fuel costs to the 20-year model.
- Perform protection coordination, short-circuit, grounding, anti-islanding, islanding, and synchronization studies.
- Recalculate NPV, IRR, and payback after validating inputs; include scenario and combined sensitivity tests.
- Use a documented Indian grid-emission factor and include diesel and lifecycle effects in the environmental calculation.

14. Conclusion

The preliminary design combines 350 kWp PV, a 250 kWh BESS, and a 150 kVA diesel generator for a representative industrial load. The model estimates 750 MWh/year of production-period demand and 511 MWh/year of PV generation, with approximately 375.1 MWh/year direct PV utilization under a synthetic hourly profile. At an assumed effective electricity price of ₹8/kWh, this yields approximately ₹30.01 lakh/year in gross direct-PV savings. The illustrative base financial case gives CAPEX of ₹233.50 lakh, IRR of 7.51%, simple payback of 11.09 years, and NPV of approximately −₹37.44 lakh at a 10% discount rate. Thus, potential technical and environmental benefits are indicated, but financial viability is not established under the base assumptions. Detailed engineering validation and verified cost/tariff data are required before implementation.

References and Data Sources

1. Punjab State Power Corporation Limited (PSPCL), tariff schedule for FY 2026–27. Verify applicability, category, billing units, and charges against the actual facility bill.

2. Sunwave Tech, industrial solar EPC cost guide, used only as an indicative planning benchmark: https://www.sunwavetech.com/blog/solar-epc-company-india

3. DCNT Global, India commercial and industrial BESS cost guide, used only as an indicative planning benchmark: https://www.dcntglobal.com/bess-india-2026-a-ci-buyers-guide/

4. MATLAB scripts and user-provided MATLAB screenshots generated for this project; outputs are illustrative, not field measurements.

Before submission, add the exact tariff document, vendor quotations, equipment datasheets, documented grid-emission-factor source, and MATLAB version details.

Appendix A — Industrial Load MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix A - Industrial Load Profile
clc; clear; close all;
load_kW = [100 120 140 160 180 200 160 120 130 150 170 190 250 200
120 110];
hour = 6:21;
dt = 1;
daily_energy_kWh = sum(load_kW)*dt;
annual_days = 300;
annual_energy_MWh = daily_energy_kWh*annual_days/1000;
figure('Name','Representative Industrial Load Profile');
stairs(hour,load_kW,'LineWidth',1.5);
xlabel('Hour of Day'); ylabel('Demand (kW)');
title('Representative Industrial Load Profile'); grid on;
fprintf('Daily energy = %.1f kWh\n',daily_energy_kWh);
fprintf('Average operating load = %.2f kW\n',mean(load_kW));
fprintf('Peak hourly-average load = %.1f kW\n',max(load_kW));
fprintf('Annual energy = %.1f MWh/year\n',annual_energy_MWh);
```

Appendix B — Solar PV MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix B - Solar PV and Load Profile
clc; clear; close all;
load_kW = [100 120 140 160 180 200 160 120 130 150 170 190 250 200
120 110];
load_hours = 6:21;
PV_kWp = 350; PSH = 5; PR = 0.80;
daily_PV_kWh = PV_kWp*PSH*PR;
solar_factor = [0.05 0.15 0.30 0.50 0.70 0.85 0.95 1.00 0.95 0.80
0.60 0.35 0.10];
solar_hours = 6:18;
PV_kW = solar_factor*daily_PV_kWh/sum(solar_factor);
figure('Name','Industrial Load and Solar PV Generation');
stairs(load_hours,load_kW,'LineWidth',1.5); hold on;
stairs(solar_hours,PV_kW,'LineWidth',1.5); hold off;
xlabel('Hour of Day'); ylabel('Power (kW)');
title('Industrial Load and Solar PV Generation');
legend('Industrial Load','Solar PV Output','Location','best'); grid
on;
fprintf('PV energy estimate = %.1f kWh/day; %.1f MWh/
year\n',daily_PV_kWh,daily_PV_kWh*365/1000);
```

Appendix C — Battery Sizing and SOC MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix C - Battery Sizing and SOC During Critical-Load Backup
clc; clear; close all;
Pcritical_kW = 100; duration_h = 2;
DoD = 0.90; eta_discharge = 0.95;
Erequired_kWh = Pcritical_kW*duration_h;
Enominal_kWh = Erequired_kWh/(DoD*eta_discharge);
Ebattery_kWh = 250; SOC0 = 100; SOCmin = 10; Pdischarge_kW = 100;
dt = 0.05; t = 0:dt:duration_h;
SOC = zeros(size(t)); SOC(1)=SOC0;
for k=2:length(t)
    deltaSOC = (Pdischarge_kW*dt/(Ebattery_kWh*eta_discharge))*100;
    SOC(k)=max(SOCmin,SOC(k-1)-deltaSOC);
end
figure('Name','Battery SOC During Critical Load Backup');
plot(t,SOC,'LineWidth',1.5);
xlabel('Time (h)'); ylabel('State of Charge (%)');
title('Battery State of Charge During Critical-Load Backup'); grid
on;
fprintf('Useful AC energy required = %.2f kWh\n',Erequired_kWh);
fprintf('Nominal capacity required = %.2f kWh\n',Enominal_kWh);
fprintf('Candidate battery AC-equivalent energy = %.2f
kWh\n',Ebattery_kWh*DoD*eta_discharge);
```

Appendix D — Diesel Generator Sizing MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix D - Diesel Generator Sizing
clc; clear; close all;
Pcritical_kW = 100; PF = 0.90; target_loading = 0.80;
Sload_kVA = Pcritical_kW/PF;
Srequired_kVA = Sload_kVA/target_loading;
Sgenerator_kVA = 150;
Pgenerator_kW = Sgenerator_kVA*PF;
loading_percent = Pcritical_kW/Pgenerator_kW*100;
figure('Name','Critical Load and Candidate Generator Rating');
bar([Pcritical_kW Pgenerator_kW]);
set(gca,'XTickLabel',{'Critical Load (kW)','DG Active Capability
(kW)'});
ylabel('Power (kW)'); title('Critical-Load Requirement and Candidate
Generator Rating'); grid on;
fprintf('Critical-load apparent power = %.2f kVA\n',Sload_kVA);
fprintf('Generator rating required at target loading = %.2f
kVA\n',Srequired_kVA);
fprintf('Candidate generator active capability = %.2f
kW\n',Pgenerator_kW);
fprintf('Approximate active-power loading = %.1f%%
\n',loading_percent);
```

Appendix E — Preliminary Microgrid Power-Balance MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix E - Preliminary Microgrid Power Balance
clc; clear; close all;
load_kW = [100 120 140 160 180 200 160 120 130 150 170 190 250 200
120 110];
hour = 6:21;
dailyPV_kWh = 350*5*0.80;
factor = [0.05 0.15 0.30 0.50 0.70 0.85 0.95 1.00 0.95 0.80 0.60 0.35
0.10];
PV_kW = zeros(size(load_kW));
PV_kW(1:13) = factor*dailyPV_kWh/sum(factor);
directPV = min(load_kW,PV_kW);
surplus = max(PV_kW-load_kW,0);
deficit = max(load_kW-PV_kW,0);
figure('Name','Preliminary Microgrid Power Balance');
stairs(hour,load_kW,'LineWidth',1.5); hold on;
stairs(hour,PV_kW,'LineWidth',1.5);
stairs(hour,deficit,'LineWidth',1.5);
stairs(hour,surplus,'LineWidth',1.5); hold off;
xlabel('Hour of Day'); ylabel('Power (kW)');
title('Preliminary Microgrid Power Balance');
legend('Load','PV','Grid Deficit Before Battery','Potential PV
Surplus','Location','best');
grid on;
fprintf('Direct PV use for synthetic operating day = %.2f
kWh\n',sum(directPV));
fprintf('Potential PV surplus during modelled hours = %.2f
kWh\n',sum(surplus));
```

Appendix F — Integrated Microgrid Normal-Operation MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix F - Integrated Microgrid Normal Operation
clc; clear; close all;
load_kW = [100 120 140 160 180 200 160 120 130 150 170 190 250 200
120 110];
hour = 6:21; dt=1;
dailyPV_kWh = 350*5*0.80;
factor = [0.05 0.15 0.30 0.50 0.70 0.85 0.95 1.00 0.95 0.80 0.60 0.35
0.10];
PV_kW=zeros(size(load_kW));
PV_kW(1:13)=factor*dailyPV_kWh/sum(factor);
battery_kWh=250; SOCmin=10; SOCmax=100; SOC=SOCmax;
eta_ch=0.95; eta_dis=0.95; SOC_trace=zeros(size(hour));
grid_import=zeros(size(hour)); battery_charge=zeros(size(hour));
for k=1:length(hour)
    direct=min(load_kW(k),PV_kW(k));
    surplus=max(PV_kW(k)-load_kW(k),0);
    deficit=max(load_kW(k)-PV_kW(k),0);
    % Simplified policy: charge only from PV surplus; preserve
reserve and do not discharge in normal operation.
    available_storage=(SOCmax-SOC)/100*battery_kWh;
    charge=min(surplus,available_storage/(eta_ch*dt));
    battery_charge(k)=charge;
    SOC=min(SOCmax,SOC+(charge*dt*eta_ch/battery_kWh)*100);
    grid_import(k)=deficit;
    SOC_trace(k)=SOC;
end
figure('Name','Microgrid Energy Management During Normal Operation');
stairs(hour,load_kW,'LineWidth',1.5); hold on;
stairs(hour,PV_kW,'LineWidth',1.5);
stairs(hour,grid_import,'LineWidth',1.5); hold off;
xlabel('Hour of Day'); ylabel('Power (kW)');
title('Microgrid Energy Management During Normal Operation');
legend('Load','PV Output','Grid Import','Location','best'); grid on;
figure('Name','Battery State of Charge During Normal Operation');
plot(hour,SOC_trace,'-o','LineWidth',1.5);
xlabel('Hour of Day'); ylabel('SOC (%)');
title('Battery State of Charge During Normal Operation'); grid on;
```

Appendix G — Grid-Outage Simulation MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix G - Simplified Grid-Outage Scenarios
clc; clear; close all;
Pcritical_kW=100; Ebattery_kWh=250; eta_dis=0.95;
SOC0=100; SOCmin=10; dt=0.01;
% Scenario 1: battery-only, two-hour outage
t1=0:dt:2; SOC1=zeros(size(t1)); SOC1(1)=SOC0;
for k=2:length(t1)
    SOC1(k)=max(SOCmin,SOC1(k-1)-(Pcritical_kW*dt/
(Ebattery_kWh*eta_dis))*100);
end
figure('Name','Battery Only Backup During Grid Outage');
plot(t1,SOC1,'LineWidth',1.5);
xlabel('Time (h)'); ylabel('Battery SOC (%)');
title('Battery-Only Backup During Grid Outage'); grid on;
% Scenario 2: battery supports critical load during DG start delay,
then DG takes over.
startup_delay_h=0.25; t2=0:dt:2;
SOC2=SOC0*ones(size(t2)); Pbat=zeros(size(t2)); Pdg=zeros(size(t2));
for k=1:length(t2)
    if t2(k)<startup_delay_h
        Pbat(k)=Pcritical_kW;
        SOC2(k)=max(SOCmin,SOC0-(Pcritical_kW*t2(k)/
(Ebattery_kWh*eta_dis))*100);
    else
        Pdg(k)=Pcritical_kW;
        SOC2(k)=max(SOCmin,SOC0-(Pcritical_kW*startup_delay_h/
(Ebattery_kWh*eta_dis))*100);
    end
end
figure('Name','Battery and Diesel Generator During Prolonged
Outage');
yyaxis left; plot(t2,SOC2,'LineWidth',1.5); ylabel('Battery SOC
(%)');
yyaxis right; plot(t2,Pbat,'LineWidth',1.5); hold on;
plot(t2,Pdg,'LineWidth',1.5); hold off;
ylabel('Power (kW)'); xlabel('Time (h)');
title('Battery and Diesel Generator During Prolonged Outage');
legend('SOC','Battery Output','Diesel Output','Location','best');
grid on;
```

Appendix H — Annual Energy and Cost Comparison MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix H - Preliminary Annual Energy and Cost Comparison
clc; clear; close all;
load_kW=[100 120 140 160 180 200 160 120 130 150 170 190 250 200 120
110];
daily_load_kWh=sum(load_kW); operating_days=300;
annual_load_kWh=daily_load_kWh*operating_days;
PV_kWp=350; PSH=5; PR=0.80;
daily_PV_kWh=PV_kWp*PSH*PR; annual_PV_kWh=daily_PV_kWh*365;
factor=[0.05 0.15 0.30 0.50 0.70 0.85 0.95 1.00 0.95 0.80 0.60 0.35
0.10];
PV_kW=factor*daily_PV_kWh/sum(factor);
load_daylight=load_kW(1:length(PV_kW));
directPV_day=sum(min(PV_kW,load_daylight));
annual_directPV=directPV_day*operating_days;
tariff=8;
baseline_cost=annual_load_kWh*tariff;
grid_after=max(annual_load_kWh-annual_directPV,0);
cost_after=grid_after*tariff;
saving=baseline_cost-cost_after;
fprintf('Daily load = %.1f kWh; annual load = %.1f
MWh\n',daily_load_kWh,annual_load_kWh/1000);
fprintf('Annual PV estimate = %.1f MWh\n',annual_PV_kWh/1000);
fprintf('Annual direct PV use = %.1f MWh\n',annual_directPV/1000);
fprintf('Baseline cost = Rs %.2f lakh/year\n',baseline_cost/1e5);
fprintf('Cost after direct PV = Rs %.2f lakh/year\n',cost_after/1e5);
fprintf('Gross direct-PV saving = Rs %.2f lakh/year\n',saving/1e5);
figure('Name','Preliminary Annual Grid-Energy Comparison');
bar([annual_load_kWh grid_after]/1000);
set(gca,'XTickLabel',{'Baseline Grid Energy','Grid Energy After
Direct PV'});
ylabel('Annual Energy (MWh)'); title('Preliminary Annual Grid-Energy
Comparison'); grid on;
```

Appendix I — 20-Year Financial Analysis MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix I - 20-Year Financial Analysis (Illustrative)
clc; clear; close all;
project_life=20; discount_rate=0.10;
PV_CAPEX=126; BESS_CAPEX=62.5; DG_CAPEX=15; integration_CAPEX=20;
other_CAPEX=10; % INR lakh
initial_CAPEX=PV_CAPEX+BESS_CAPEX+DG_CAPEX+integration_CAPEX+other_CA
PEX;
annual_PV_saving=30.01; annual_OPEX=5;
annual_incremental_fuel_cost=0; other_benefits=0;
annual_net=annual_PV_saving+other_benefits-annual_OPEX-
annual_incremental_fuel_cost;
replacement=zeros(project_life,1); replacement(10)=BESS_CAPEX*0.70;
cf=zeros(project_life+1,1); cf(1)=-initial_CAPEX;
for y=1:project_life, cf(y+1)=annual_net-replacement(y); end
years=(0:project_life)';
NPV=sum(cf./(1+discount_rate).^years);
npvfun=@(r) sum(cf./(1+r).^years);
IRR=NaN;
if npvfun(-0.99)*npvfun(5)<0, IRR=fzero(npvfun,[-0.99 5]); end
cumcf=cumsum(cf); idx=find(cumcf>=0,1);
if isempty(idx), simple_payback=NaN; else
    yr=idx-1;
    if yr==0, simple_payback=0;
    else, simple_payback=(yr-1)+abs(cumcf(idx-1))/cf(idx); end
end
dcf=cf./(1+discount_rate).^years; cumdcf=cumsum(dcf);
didx=find(cumdcf>=0,1);
if isempty(didx), discounted_payback=NaN; else,
discounted_payback=didx-1; end
fprintf('Initial CAPEX = Rs %.2f lakh\n',initial_CAPEX);
fprintf('Annual gross PV saving = Rs %.2f lakh\n',annual_PV_saving);
fprintf('Annual OPEX = Rs %.2f lakh\n',annual_OPEX);
fprintf('Annual net before replacement = Rs %.2f lakh\n',annual_net);
fprintf('NPV at %.1f%% = Rs %.2f lakh\n',discount_rate*100,NPV);
fprintf('IRR = %.2f%%\n',IRR*100);
fprintf('Simple payback = %.2f years\n',simple_payback);
if isnan(discounted_payback), fprintf('Discounted payback not reached
within project life\n');
```

```
else, fprintf('Discounted payback = %.2f
years\n',discounted_payback); end
figure('Name','20-Year Industrial Microgrid Cash Flow');
bar(years,cf); xlabel('Project Year'); ylabel('Cash Flow (INR
lakh)'); title('20-Year Industrial Microgrid Cash Flow'); grid on;
figure('Name','Cumulative Cash Flow Over 20 Years');
plot(years,cumcf,'-o','LineWidth',1.5); hold on; yline(0,'--'); hold
off; xlabel('Project Year'); ylabel('Cumulative Cash Flow (INR
lakh)'); title('Cumulative Cash Flow Over 20 Years'); grid on;
figure('Name','Discounted Cumulative Cash Flow Over 20 Years');
plot(years,cumdcf,'-o','LineWidth',1.5); hold on; yline(0,'--'); hold
off; xlabel('Project Year'); ylabel('Discounted Cumulative Cash Flow
(INR lakh)'); title('Discounted Cumulative Cash Flow Over 20 Years');
grid on;
```

Appendix J — Sensitivity Analysis MATLAB Code

Complete MATLAB script. The script is intended as a reproducible starting point; where model assumptions are illustrative, they are labelled in the comments.

```
%% Appendix J - Sensitivity Analysis
clc; clear; close all;
N=20; r0=0.10; CAPEX=233.50; pvSave=30.01; OPEX=5.00; repl=43.75;
years=(0:N)'; baseCF=zeros(N+1,1); baseCF(1)=-CAPEX;
for y=1:N, baseCF(y+1)=pvSave-OPEX; end
baseCF(11)=baseCF(11)-repl;
npv=@(cf,r) sum(cf./(1+r).^years);
baseNPV=npv(baseCF,r0);
fprintf('Base CAPEX %.2f lakh; PV saving %.2f lakh/year; OPEX %.2f
lakh/year; NPV %.2f lakh\n',CAPEX,pvSave,OPEX,baseNPV);
% Tariff
tariffs=[6 8 10 12]; tariffNPV=zeros(size(tariffs));
for i=1:numel(tariffs)
    cf=baseCF; annual=pvSave*(tariffs(i)/8)-OPEX;
    for y=1:N, cf(y+1)=annual; end
    cf(11)=cf(11)-repl; tariffNPV(i)=npv(cf,r0);
end
figure; plot(tariffs,tariffNPV,'-o','LineWidth',1.5); hold on;
yline(0,'--'); hold off; xlabel('Assumed Electricity Tariff (INR/
kWh)'); ylabel('NPV (INR lakh)'); title('NPV Sensitivity to
Electricity Tariff'); grid on;
% CAPEX
factors=[0.8 1 1.2]; capNPV=zeros(size(factors));
for i=1:numel(factors), cf=baseCF; cf(1)=-CAPEX*factors(i);
capNPV(i)=npv(cf,r0); end
figure; plot(factors*100,capNPV,'-o','LineWidth',1.5); hold on;
yline(0,'--'); hold off; xlabel('Initial CAPEX (% of Base Case)');
ylabel('NPV (INR lakh)'); title('NPV Sensitivity to Initial Capital
Cost'); grid on;
% OPEX
ox=[3 5 7]; oxNPV=zeros(size(ox));
for i=1:numel(ox)
 cf=baseCF; for y=1:N, cf(y+1)=pvSave-ox(i); end; cf(11)=cf(11)-repl;
oxNPV(i)=npv(cf,r0);
end
figure; plot(ox,oxNPV,'-o','LineWidth',1.5); hold on; yline(0,'--');
hold off; xlabel('Annual OPEX (INR lakh/year)'); ylabel('NPV (INR
lakh)'); title('NPV Sensitivity to Annual Operating Expenses'); grid
on;
```

```
% PV savings
pf=[0.8 1 1.1]; pfNPV=zeros(size(pf));
for i=1:numel(pf)
 cf=baseCF; for y=1:N, cf(y+1)=pvSave*pf(i)-OPEX; end; cf(11)=cf(11)-
repl; pfNPV(i)=npv(cf,r0);
end
figure; plot(pf*100,pfNPV,'-o','LineWidth',1.5); hold on;
yline(0,'--'); hold off; xlabel('Direct PV Saving (% of Base Case)');
ylabel('NPV (INR lakh)'); title('NPV Sensitivity to PV Energy
Utilization'); grid on;
% Discount rate
rates=[0.06 0.08 0.10 0.12]; rateNPV=zeros(size(rates));
for i=1:numel(rates), rateNPV(i)=npv(baseCF,rates(i)); end
figure; plot(rates*100,rateNPV,'-o','LineWidth',1.5); hold on;
yline(0,'--'); hold off; xlabel('Discount Rate (%)'); ylabel('NPV
(INR lakh)'); title('NPV Sensitivity to Discount Rate'); grid on;
fprintf('\nTariff sensitivity:\n'); for i=1:numel(tariffs),
fprintf('Rs %.2f/kWh -> NPV Rs %.2f lakh\n',tariffs(i),tariffNPV(i));
end
fprintf('\nCAPEX sensitivity:\n'); for i=1:numel(factors),
fprintf('%.0f%% CAPEX -> NPV Rs %.2f
lakh\n',100*factors(i),capNPV(i)); end
fprintf('\nOPEX sensitivity:\n'); for i=1:numel(ox), fprintf('Rs %.2f
lakh/year -> NPV Rs %.2f lakh\n',ox(i),oxNPV(i)); end
fprintf('\nPV savings sensitivity:\n'); for i=1:numel(pf),
fprintf('%.0f%% PV savings -> NPV Rs %.2f
lakh\n',100*pf(i),pfNPV(i)); end
fprintf('\nDiscount sensitivity:\n'); for i=1:numel(rates),
fprintf('%.0f%% discount -> NPV Rs %.2f
lakh\n',100*rates(i),rateNPV(i)); end
```